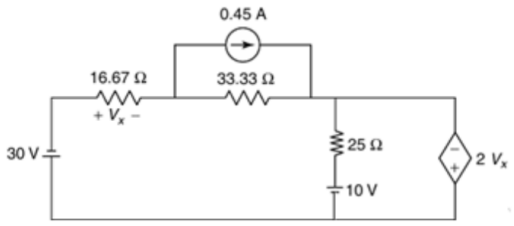
THIRD SEMESTER B.TECH. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - DECEMBER 2024/JANUARY 2025

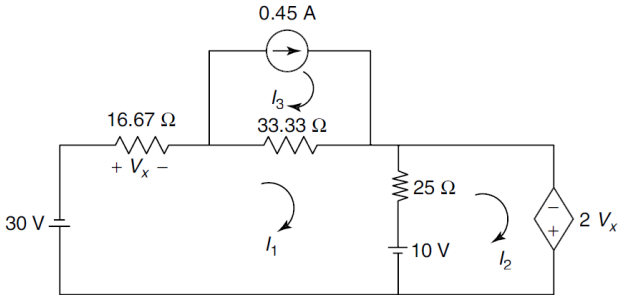
SUBJECT: ECE 2122- NETWORK ANALYSIS

NETWORK ANALYSIS

1A) Find the voltage *Vx* in the network shown below figure. (4)



Solution:



$$V_x = 16.67\, I_1 \quad \ldots(i)$$ 1 mark

Applying KVL to Mesh 1,

$$-30 - 16.67\, I_1 - 33.33(I_1 - I_3) - 25(I_1 - I_2) - 10 = 0$$

$$-30 - 16.67\, I_1 - 33.33\, I_1 + 33.33\, I_3 - 25\, I_1 + 25\, I_2 - 10 = 0$$

$$-75\, I_1 + 25\, I_2 + 33.33\, I_3 = 40 \quad \ldots(ii)$$ 1 mark

Applying KVL to Mesh 2,

$$10 - 25(I_2 - I_1) + 2\, V_x = 0$$

$$10 - 25(I_2 - I_1) + 2(16.67\, I_1) = 0$$

$$10 - 25\, I_2 + 25\, I_1 + 33.34\, I_1 = 0$$

$$58.34\, I_1 - 25\, I_2 = -10 \quad \ldots(iii)$$ 1 mark

For Mesh 3,

$$I_3 = 0.45 \quad \ldots(iv)$$

Solving Eqs (ii), (iii) and (iv),

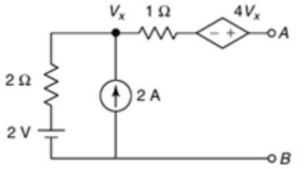
$$I_1 = -0.9 \text{ A}$$

$$I_2 = -1.7 \text{ A}$$

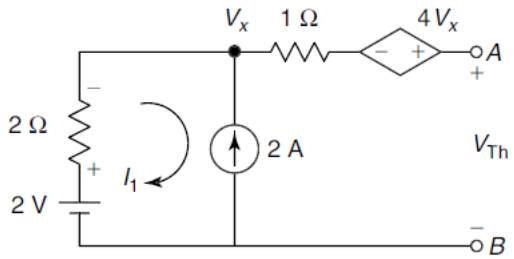
$$I_3 = 0.45 \text{ A}$$

$$V_x = 16.67\, I_1 = 16.67\,(-0.9) = -15 \text{ V} \quad \ldots(v)$$ 1 mark

1B) Obtain the Thevenin equivalent network of figure shown below with respect to terminals A and B. (4)



Solution



Step I Calculation of V_{Th}

$$2-2I_1-V_x=0$$
$$V_x=2-2I_1$$

1 mark

For Mesh 1,

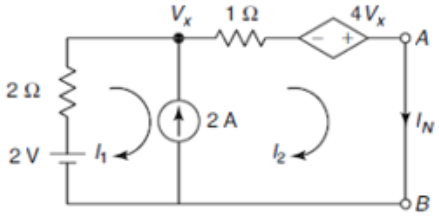
$$I_1=-2\text{ A}$$
$$V_x=2-2(-2)=6\text{ V}$$

Writing the V_{Th} equation,

$$2-2I_1-0+4V_x-V_{Th}=0$$
$$2-2(-2)-0+4(6)-V_{Th}=0$$
$$V_{Th}=30\text{ V}$$

1 mark

Step II Calculation of I_N



$$V_x=2-2I_1 \qquad \ldots(i)$$

Meshes 1 and 2 will form a supermesh,
Writing current equation for the supermesh

$$I_2-I_1=2 \qquad \ldots(ii)$$

Applying KVL to the outer path of the supermesh,

$$2-2I_1-1I_2+4V_x=0$$
$$2-2I_1-I_2+4(2-2I_1)=0$$
$$10I_1+I_2=10$$

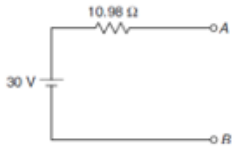
Solving Eqs (ii) and (iii),

$$I_1=0.73\text{ A}$$
$$I_2=2.73\text{ A}$$
$$I_N=I_2=2.73\text{ A}$$

1 mark

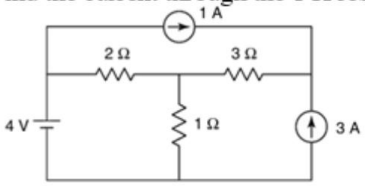
Step III Calculation of R_{Th}

$$R_{Th}=\frac{V_{Th}}{I_N}=\frac{30}{2.73}=10.98\ \Omega$$



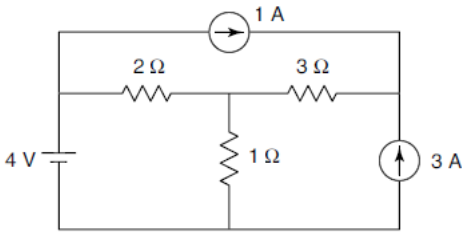
1 mark

1C) Find the current through the 1 Ω resistor using superposition theorem in network shown below.

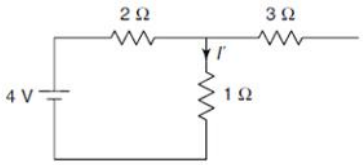


(2)

Solution:

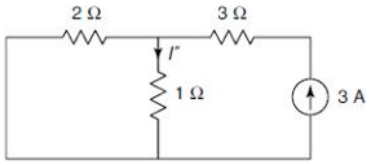


Step I When the 4 V source is acting alone



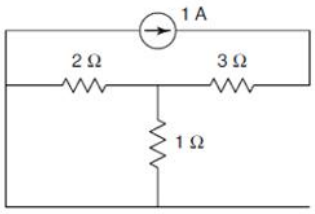
$$I' = \frac{4}{2+1} = 1.33 \text{ A}(\downarrow)$$

Step II When the 3 A source is acting alone
By current-division rule,



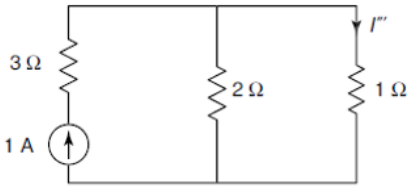
$$I'' = 3 \times \frac{2}{1+2} = 2 \text{ A } (\downarrow)$$

Step III When the 1 A source is acting alone



1 mark

Redrawing the network
By current-division rule,



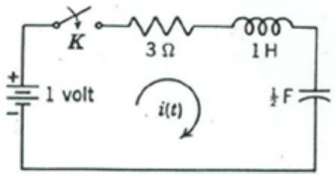
$$I''' = 1 \times \frac{2}{2+1} = 0.66 \text{ A}(\downarrow)$$

Step IV By superposition theorem,

$$I = I' + I'' + I''' = 1.33 + 2 + 0.66 = 4 \text{ A } (\downarrow)$$

1 mark

2A) In the network shown in figure, the switch K is closed at t = 0. For the numerical values given, find complementary, particular and total response i(t) using differential equation. Also draw the current response as a function of time.



(5)

Solution:

Apply KVL For t ≥ 0,

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = V \qquad (1)$$

Differentiating and using numerical values for R, L, and C

$$\frac{d^2i}{dt^2} + 3\frac{di}{dt} + 2i = 0 \qquad (2)$$

$s^2 + 3s + 2 = 0$

The roots of the above equation by quadratic formula,

$s_1 = -1$; $s_2 = -2$

The general solution is: $i(t) = i_1 + i_2$

$i(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t} = i(t) = k_1 e^{-t} + k_2 e^{-2t}$ (3) ---------- (2 Marks)

To find k values: From equation 1

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = V \quad (1)$$

If the switch K is closed at t = 0, then $i(0^+) = 0$, because current cannot change instantaneously in the inductor. In Equation 1, the second and third voltage terms are zero at the instant of switching, $Ri(0+)$ being zero because $i(0^+) = 0$ and $(1/C)\int i\,dt$ being zero because it is the initial voltage across the capacitor.

Hence,

$$\frac{di}{dt}(0+) = \frac{V}{L} = 1\ amp/sec \quad (4)$$

At $t = 0^+$, $i(0) = i(0^+) = 0$, From equation 3, $i(t) = k_1 e^{-t} + k_2 e^{-2t}$ (3)

$k_1 + k_2 = 0$

$i(t) = k_1 e^{-t} + k_2 e^{-2t}$ (3)

Differentiating equation 3,

$di/dt = -k_1 e^{-t} - 2k_2 e^{-2t}$

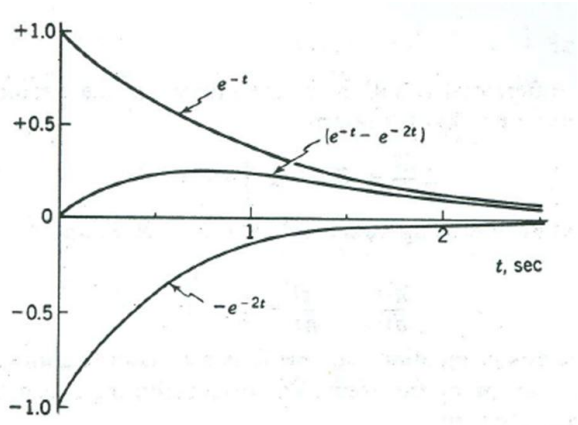
$di/dt(0+) = -k_1 - 2k_2 = 1$ (and equating with equation 4)

We have,
$k_1 + k_2 = 0$
$-k_1 - 2k_2 = 1$, solve this we get

$k_1 = 1$; $k_2 = -1$

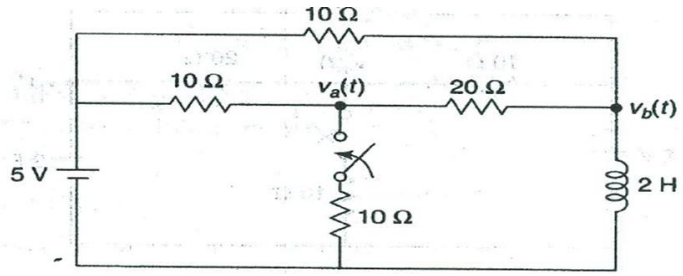
Substitute in Equation (3): $i(t) = (e^{-t} - e^{-2t})$; $t \geq 0$

---------- (2 Marks)



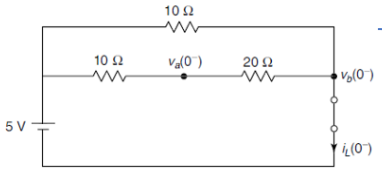
---------- (1 Mark)

2B. In the network shown in Figure, a steady state is reached with the switch open. At t = 0, the switch is closed. For the element values given, determine the values of $v_a(0^-)$, $v_b(0^-)$, $v_a(0^+)$ and $v_b(0^+)$. **(3 Marks)**



Solution:

At t = 0⁻, the switch is open and the network attains steady-state condition. Hence, the inductor acts as a short circuit. The equivalent network shown below,



$$i_L(0^-) = \frac{5}{(10 \,||\, 30)} = \frac{5}{7.5} = \frac{2}{3}\text{ A}$$

$$v_b(0^-) = 0$$

Writing the KCL at node 'a', then find $v_a(0^-)$:

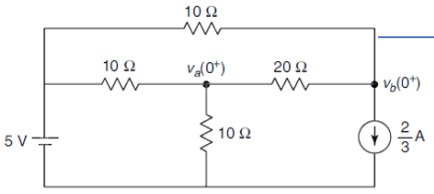
$$v_a(0^-) = 5 \times \frac{20}{30} = 3.33\text{ V}$$

---------- (1Mark)

- Note: If a current of value I_0 flows through the inductor at t = 0⁻, the inductor can be regarded as a current source of I_0 ampere at t = 0⁺.

Hence, At t = 0⁺, the inductor acts as a current source of 2/3 A.

At t = 0⁺, the switch is in closed state and the network is shown in Fig. below,



$$i_L(0^+) = \frac{2}{3}\text{ A}$$

Writing the KCL equations at node 'a'

$$\frac{v_a(0^+)-5}{10} + \frac{v_a(0^+)}{10} + \frac{v_a(0^+)-v_b(0^+)}{20} = 0$$

Writing the KCL equations at node 'b'

$$\frac{v_b(0^+)-v_a(0^+)}{20} + \frac{v_b(0^+)-5}{10} + \frac{2}{3} = 0$$

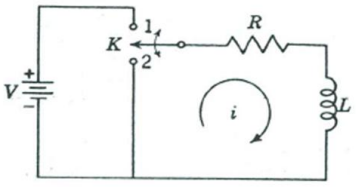
Solving these two Equations,

$$v_a(0^+) = 1.9\text{ V}$$

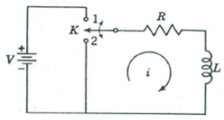
$$v_b(0^+) = -0.477\text{ V}$$

---------- (2 Marks)

2C. In the network shown in Figure, the switch K is moved from position 1 to position 2 at t = 0, a steady state current having previously been established in the RL circuit. Find the general solution for the current *i*. **(2 Marks)**



Solution:



Given Circuit

After the switching has taken place ($t \geq 0$), the Kirchhoff voltage equation is,

$v(t) = L\frac{di}{dt}$

$$L\frac{di}{dt} + Ri = 0$$

This is a first-order linear differential equation with constant coefficients.

Rearranging the above equation,

$$\frac{di}{i} = -\frac{R}{L}\mathrm{dt}$$

Integrating both sides

$$\ln i = -\frac{R}{L}\mathrm{t} + K$$

---------- (1 Mark)

To simplify the above equation, *Let* constant $K = \ln k$ and we know that $\ln e^x = x$, or $\log_{10} 10^x = x$.

Using above the equation 1 can be written as,

$$\ln i = ln\, e^{\frac{-Rt}{L}} + \ln k \qquad (2)$$

$\ln y + \ln z = \ln yz$

$$\ln i = \ln(ke^{\frac{-Rt}{L}})$$

Taking antilogarithm,

$$i = ke^{\frac{-Rt}{L}}$$

---------- (1 Mark)

3A) A 10 Hz symmetrical square wave whose peak-to-peak amplitude is 2 V is fed to an amplifier. Calculate and sketch the output waveform if the lower 3-dB frequency is 0.3 Hz. (4)

Solution:

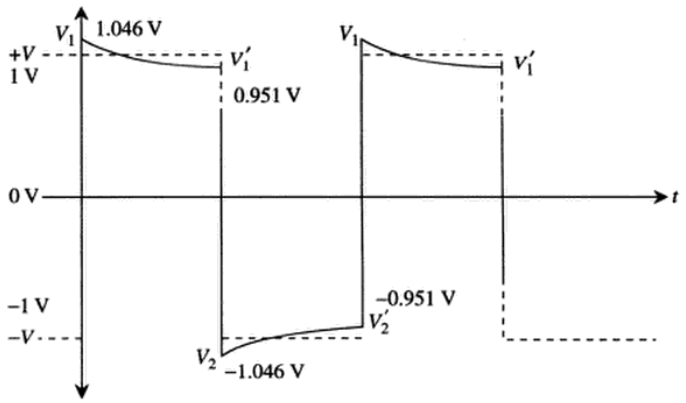
$$V_1 = \frac{V}{1 + e^{-T/2\tau}} = \frac{2}{1 + e^{-0.05/0.53}} = 1.046\,\text{V}$$

$$V_1' = V_1 e^{-T/2\tau} = 1.046 e^{-0.05/0.53} = 0.951\,\text{V}$$

$$V_1 = -V_2 \quad \text{and} \quad V_1' = -V_2'$$

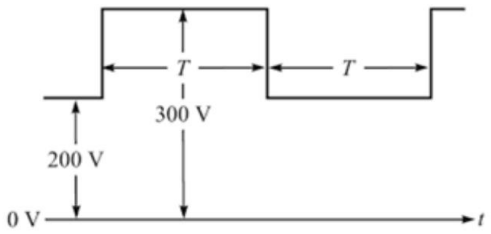
$$V_1 = |\,V_2\,| = 1.046\,\text{V} \qquad V_1' = |\,V_2'\,| = 0.951\,\text{V}$$

(3 M)



(1 M)

3B) The square wave shown in the figure is fed to an RC integrating circuit. Compute and plot the output waveform under steady-state conditions, if RC = 2.5T



(4)

Solution:

$$V_1 = 300 - (300 - V_2)e^{-T/2.5T} = 300 - (300 - V_2) \times 0.670 = 99 + 0.67V_2$$

$$V_2 = 200 - (200 - V_1)e^{-T/2.5T} = 200 - (200 - V_1) \times 0.670 = 66 + 0.67V_1$$

Solving the above equations for V_1 and V_2,

$$V_1 = 99 + 0.67(66 + 0.67\ V_1)$$

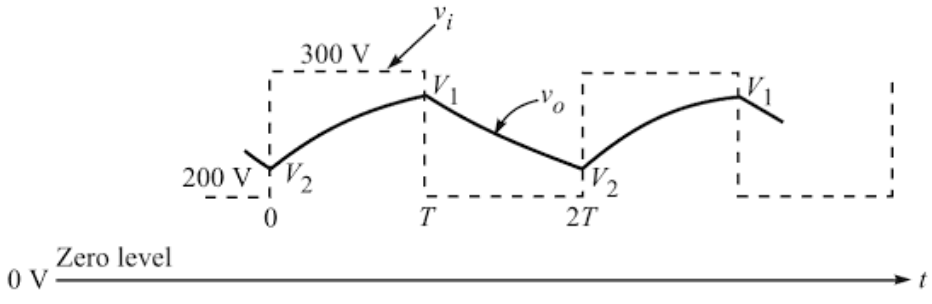
$$V_1 = 143.22/0.5511 = 259.88\ \text{V}$$

$$V_2 = 66 + 0.67(259.88) = 240.12\ \text{V}$$

The maximum and minimum values of output are

$$V_1 = 259.88\ \text{V} \quad \text{and} \quad V_2 = 240.12\ \text{V}$$

(3 M)



(1 M)

3C) Prove that a High-pass RC Circuit is a Differentiator. (2)

$$v_i = \frac{1}{C}\int idt + iR$$

$$v_i = \frac{1}{C}\int idt \quad or \quad v_i = \frac{1}{\tau}\int v_o dt \ \text{(since } i = V_o/R\text{)}$$

(1M)

Differentiating: $\frac{dv_i}{dt} = \frac{v_o}{\tau}$ $\qquad v_o = \tau\frac{dv_i}{dt}$ $\qquad v_o \propto \frac{dv_i}{dt}$

It can be seen that the output is proportional to the differential of the input signal (1M)

4A) A general network consisting of a resistor and an inductor is shown in Fig. 4A. The switch is closed at time t = 0. Using Laplace transform, obtain the expression for i(t) for t > 0.

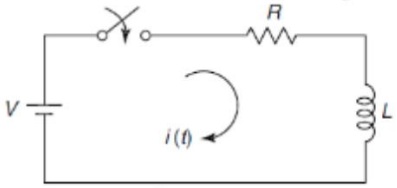
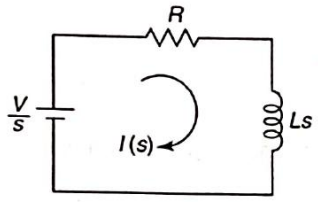


Fig: 4A

(4)

Soln:

For t > 0, the transformed network is



Applying KVL to the mesh,

$$\frac{V}{s} - RI(s) - Ls\,I(s) = 0$$

$$I(s) = \frac{\frac{V}{L}}{s\left(s+\frac{R}{L}\right)}$$

By partial-fraction expansion,

$$I(s) = \frac{A}{s} + \frac{B}{s+\frac{R}{L}}$$

$$A = sI(s)\big|_{s=0} = s \times \left.\frac{\frac{V}{L}}{s\left(s+\frac{R}{L}\right)}\right|_{s=0} = \frac{V}{R}$$

$$B = \left(s+\frac{R}{L}\right)I(s)\Big|_{s=-\frac{R}{L}} = \left(s+\frac{R}{L}\right) \times \left.\frac{\frac{V}{L}}{s\left(s+\frac{R}{L}\right)}\right|_{s=-\frac{R}{L}} = -\frac{V}{R}$$

$$I(s) = \frac{\frac{V}{R}}{s} + \frac{\left(-\frac{V}{R}\right)}{s+\frac{R}{L}}$$

Taking the inverse Laplace transform,

$$i(t) = \frac{V}{R} - \frac{V}{R}e^{-\frac{R}{L}t}$$

$$= \frac{V}{R}\left[1 - e^{-\frac{R}{L}t}\right] \quad \text{for } t > 0$$

4B) In the network shown in Fig. 4B, the switch is opened at t = 0. Steady-state condition is achieved before t = 0. Using Laplace transform, calculate i(t).

4 marks

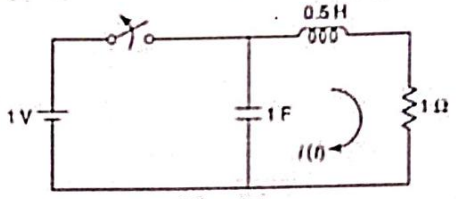
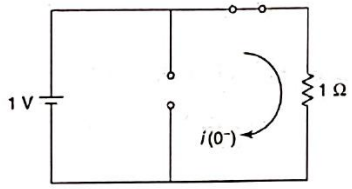


Fig: 4B

Soln:

At t = 0-, the network is



— 0.5

At t = 0-, the switch is closed, and steady-state condition is achieved. Hence, the capacitor acts as an open circuit and the inductor acts as a short circuit.

$$v_c(0^-) = 1 \text{ V}$$
$$i(0^-) = 1 \text{ A}$$

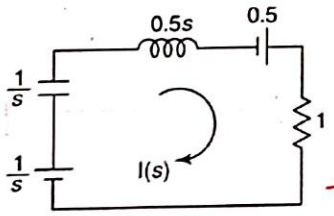
— 0.5

Since current through the inductor and voltage across the capacitor cannot change instantaneously,

$$v_c(0^+) = 1 \text{ V}$$
$$i(0^+) = 1 \text{ A}$$

— 0.5

For t > 0, the transformed circuit is



— 0.5

Applying KVL to the mesh for $t > 0$,

$$\frac{1}{s} - \frac{1}{s} I(s) - 0.5s\, I(s) + 0.5 - I(s) = 0$$

$$0.5 + \frac{1}{s} = \frac{1}{s} I(s) + 0.5s\, I(s) + I(s)$$

$$I(s)\left[1 + \frac{1}{s} + 0.5s\right] = 0.5 + \frac{1}{s}$$

$$I(s) = \frac{s+2}{s^2+2s+2} = \frac{(s+1)+1}{(s+1)^2+1} = \frac{s+1}{(s+1)^2+1} + \frac{1}{(s+1)^2+1}$$

1

Taking the inverse Laplace transform,

$$i(t) = e^{-t}\cos t + e^{-t}\sin t \qquad \text{for } t > 0$$

— 1

4C) In the network of Fig. 4C, the switch is moved from a to b, at t = 0. For t > 0, calculate v(t) using Laplace transform.

2 marks

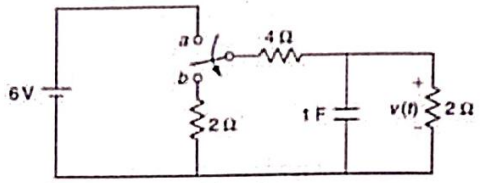
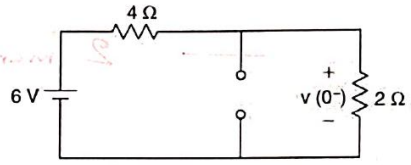


Fig: 4C

Soln:

At t = 0−, the network is



0.5

At t = 0−, steady-state condition is reached. Hence, the capacitor acts as an open circuit.

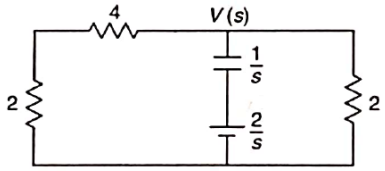
$$v(0^-) = 6 \times \frac{2}{4+2} = 2 \text{ V}$$

Since voltage across the capacitor cannot change instantaneously,

$$v(0^+) = 2 \text{ V}$$

0.5

For t > 0, the transformed network is



0.5

Applying KCL at the node for $t > 0$,

$$\frac{V(s)}{6} + \frac{V(s) - \frac{2}{s}}{\frac{1}{s}} + \frac{V(s)}{2} = 0$$

$$V(s)\left(\frac{2}{3} + s\right) = 2$$

$$V(s) = \frac{2}{s + \frac{2}{3}}$$

0.5

Taking the inverse Laplace transform,

$$v(t) = 2e^{-\frac{2}{3}t} \quad \text{for } t > 0$$

5A) Obtain the voltage transfer function of the two-port network shown in Fig. 5A

3 marks

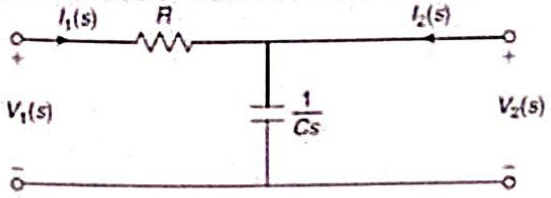


Fig: 5A

Soln:

By voltage division rule,

$$V_2(s) = V_1(s)\frac{\frac{1}{Cs}}{R+\frac{1}{Cs}} = V_1(s)\frac{1}{RCs+1} = V_1(s)\frac{\frac{1}{RC}}{s+\frac{1}{RC}}$$ — 2 marks

Voltage transfer function $$\frac{V_2(s)}{V_1(s)} = \frac{\frac{1}{RC}}{s+\frac{1}{RC}}$$ — 1 mark

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5B) Obtain the Y and Z-parameters of the network shown in Fig. 5B

4 marks

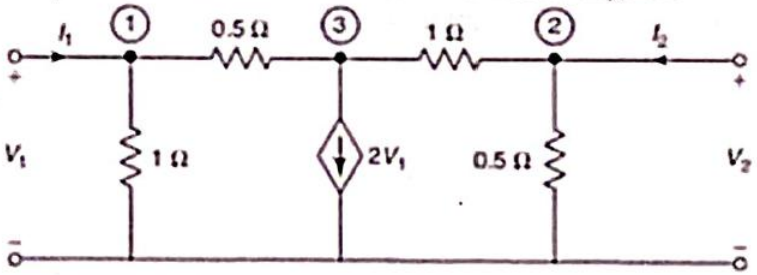


Fig: 5B

Solution Applying KCL at Node 1,

$$I_1 = \frac{V_1}{1}+\frac{V_1-V_3}{0.5}$$
$$= 3V_1 - 2V_3 \quad \ldots(i)$$

Applying KCL at Node 2,

$$I_2 = \frac{V_2}{0.5}+\frac{V_2-V_3}{1}$$
$$= 3V_2 - V_3 \quad \ldots(ii)$$

Applying KCL at Node 3,

$$\frac{V_3-V_1}{0.5}+2V_1+\frac{V_3-V_2}{1}=0$$
$$V_3 = \frac{1}{3}V_2 \quad \ldots(iii)$$

1

Substituting Eq. (iii) in Eqs (i) and (ii),

$$I_1 = 3V_1 - \frac{2}{3}V_2 \quad \ldots(iv)$$

$$I_2 = 0V_1 + \frac{8}{3}V_2 \quad \ldots(v)$$

0.5

Comparing Eqs (iv) and (v) with Y-parameter equations,

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} 3 & -\frac{2}{3} \\ 0 & \frac{8}{3} \end{bmatrix}$$

0.5

Z-parameters

$$\Delta Y = Y_{11}Y_{22} - Y_{12}Y_{21} = (3)\left(\frac{8}{3}\right) - 0 = 8$$

$$Z_{11} = \frac{Y_{22}}{\Delta Y} = \frac{\frac{8}{3}}{8} = \frac{1}{3}\Omega, \qquad Z_{12} = -\frac{Y_{12}}{\Delta Y} = \frac{\frac{2}{3}}{8} = \frac{1}{12}\Omega$$

$$Z_{21} = -\frac{Y_{12}}{\Delta Y} = \frac{\frac{2}{3}}{8} = \frac{1}{12}\Omega, \qquad Z_{22} = \frac{Y_{11}}{Y\Delta} = \frac{3}{8}\Omega$$

1.5

Hence, the Z-parameters are

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{12} \\ 0 & \frac{3}{8} \end{bmatrix}$$

0.5

5C) **Obtain the ABCD parameters for the two-port network shown in Fig. 5C**

3 marks

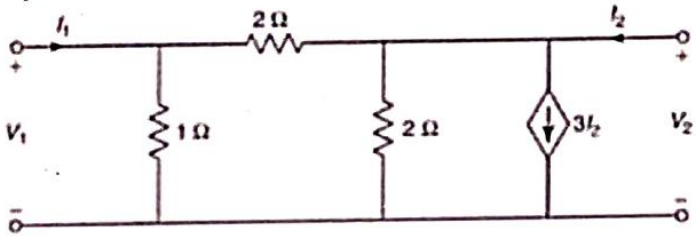
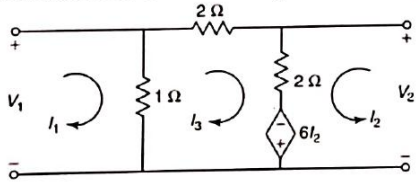


Fig: 5C

Soln:

The network is redrawn by source transformation technique



Applying KVL to Mesh 1,

$$V_1 = I_1 - I_3 \quad \ldots(i)$$

Applying KVL to Mesh 2,

$$V_2 = 2(I_2 + I_3) - 6I_2$$
$$= -4I_2 + 2I_3 \quad \ldots(ii)$$

Applying KVL to Mesh 3,

$$-(I_3 - I_1) - 2I_3 - 2(I_2 + I_3) + 6I_2 = 0$$
$$5I_3 = I_1 + 4I_2$$
$$I_3 = \frac{1}{5}I_1 + \frac{4}{5}I_2 \quad \ldots(iii)$$

1

Substituting Eq. (iii) in Eq. (i),

$$V_1 = I_1 - \frac{1}{5}I_1 - \frac{4}{5}I_2$$

$$= \frac{4}{5}I_1 - \frac{4}{5}I_2 \quad \ldots(iv)$$

Substituting Eq. (iii) in Eq. (ii),

$$V_2 = -4I_2 + 2\left(\frac{1}{5}I_1 + \frac{4}{5}I_2\right)$$

$$= \frac{2}{5}I_1 - \frac{12}{5}I_2 \quad \ldots(v)$$

0.5

Comparing Eqs (iv) and (v) with Z-parameter equations,

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & -\frac{4}{5} \\ \frac{2}{5} & -\frac{12}{5} \end{bmatrix}$$

0.5

ABCD parameters:

$$\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21} = \left(\frac{4}{5}\right)\left(-\frac{12}{5}\right) - \left(-\frac{4}{5}\right)\left(\frac{2}{5}\right) = -\frac{40}{25} = -\frac{8}{5}$$

$$A = \frac{Z_{11}}{Z_{21}} = 2 \qquad B = \frac{\Delta Z}{Z_{21}} = -4$$

$$C = \frac{1}{Z_{21}} = \frac{5}{2} \qquad D = \frac{Z_{22}}{Z_{21}} = -6$$

(1M)
